A benchmark of cell-type annotation methods for single-cell data: cost, not accuracy, distinguishes them

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Abstract

Cell-type annotation by reference mapping is among the most frequently repeated operations in single-cell analysis, yet published comparisons report accuracy far more often than the wall-clock time and memory that decide what a working scientist can actually run. We benchmarked **thirteen methods** — classical, regularized-linear, parameter-free, correlation, deep-probabilistic, prototype-VAE and foundation-model — across **six datasets** (8–86 cell types; within-dataset, cross-dataset and cross-study splits) on commodity hardware, and validated externally on Open Problems `label_projection`, whose datasets, metrics and ranking we did not choose. Accuracy among the leading methods is tightly clustered — the top four span **0.008** — while their inference cost differs by two orders of magnitude and their peak memory by 2.4×. No method leads everywhere, and neither ordering is stable: accuracy inverts as the reference grows from 3k cells to a full atlas, and cost rankings invert with the input feature budget. Because several methods are now both accurate and cheap, the useful question is not which is best but which fits a given job. We show that a low, flat inference cost makes multi-stage annotation practical on a laptop, using **actinn-jax**, a JAX reimplementation of ACTINN with a cached train-once/map-many reference: a shipped ~800-type reference with calibrated abstention hands off to a user's own focused reference (cross-study liver 0.23/0.58 → 0.72/0.86, exact/ontology), with resolution below the cell-type label and cluster-level novelty screening. The same profile annotates a whole 525,000-cell atlas in 58 s at 15 GB, on a query axis where the tuned linear pipeline does not finish. The same route builds a pan-mouse reference with no GPU, and distills **Pan-human Azimuth** into a broad-pass model matching its concordance at 6–9× its throughput. We release the reimplementation, the harness and the pre-trained references.

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1 Key Points

- Among leading annotation methods, accuracy differences are small (top four within 0.008) while predict time differs by ~260× and peak memory by 2.4×; cost, not accuracy, is what distinguishes them in practice.
- Rankings are not stable: a prototype VAE moves from worst to best as the reference grows from 3k to 49k cells, and a tuned linear pipeline that fits 7× faster than a gene-space MLP on one panel costs 2.7× more on another with a narrower feature budget.
- Inference time is flat in reference size and cardinality for a cached gene-space MLP (0.08–0.33 s from 1k to 24k reference cells), which is what makes chaining several annotation stages practical on a laptop.
- A pretrained pan-human annotator can be distilled into a fast reference using only raw counts — no GPU, no labels — matching the teacher's concordance and beating a census-built reference, at 6–9× the teacher's throughput.

- Zero-shot foundation-model labels remain the weakest option in both our benchmark and the external one; their value is in learned structure, not in their label heads.

2 Introduction

Annotating cells by mapping to a labeled reference is run constantly and rarely reported as a cost. Existing comparisons [Abdelaal 2019, Fu 2024] emphasize accuracy; the axis that decides what runs on a laptop — wall-clock and memory without a GPU — is usually absent. Foundation models [Kalfon 2025] raise accuracy in some settings but need accelerators, and their zero-shot label predictions underperform small models trained on curated references.

Concurrent work sharpens the question rather than settling it. **Pan-human Azimuth** [Sarkar et al. 2026] ships a supervised hierarchical classifier over a harmonized organism-wide typology — 8 levels, 382 leaf types, ~7M parameters over a fixed 5,055-gene panel, trained on 9.7M curated cells, with abstention *learned* rather than thresholded (expected calibration error 0.0044) — and runs on a laptop. It is better resourced than any reference we could build, and its authors reach a conclusion parallel to ours: training-data quality and organization matter as much as architecture or scale, with accuracy saturating past ~5M training cells. A purpose-built pan-human model is therefore the right thing to *start* from; the open question is what to do next, since no fixed typology can re-annotate into a user's own label set or resolve states below its own leaves.

We therefore ask a practical question: for a given annotation job on commodity hardware, which method should be run, and what does the surrounding workflow look like? The leading methods now annotate quickly, accurately, and with a usable signal on unknown cells, so the shortage is not another leaderboard but guidance on fit for purpose.

Because a benchmark written by a method's own author has a known failure mode, the comparison was constrained by construction: every method runs on **every** dataset through one harness on identical splits; the panel was chosen to include the baselines most likely to beat a small MLP, and each of them does beat it somewhere; our own classical tier is left untuned while the linear baseline is tuned; and the external validation uses a benchmark designed by others.

3 Materials and methods

actinn-jax. A dependency-light JAX reimplementation of ACTINN [Ma & Pellegrini 2020] — a 4-layer network (100/50/25 hidden units, ReLU, softmax, Adam) — replacing TensorFlow-1.x graph/session code that no longer installs on current Python and ML environments. Preprocessing is sparse-aware; a fitted reference is cached and reused across queries; prediction is chunked for atlas-scale inputs. Accuracy matches the original within repeat noise.

Panel and datasets. Thirteen methods (Supplementary Table S1) across six datasets (Supplementary Table S2): lung within-dataset and cross-dataset, liver within-dataset and cross-**study**, an 86-type blood+gut set, and PBMC — 8 to 86 cell types, spanning three generalization regimes.

Metrics. Accuracy, macro-F1, and **ontology-aware concordance**, which credits a call that is the same node, an ancestor or a descendant of the truth in the Cell Ontology. The last is required because vocabularies disagree about granularity: on our cross-dataset lung split reference and query share only 20 of 46 type names, so exact-match accuracy (~0.35 for every method) measures vocabulary mismatch rather than transfer. Concordance is reported only where both sides carry ontology ids.

Execution. Each method runs in its own environment as a separate process, because the dependency sets are mutually unsatisfiable; one driver builds each split once from a fixed seed and hands the identical pair to every method. Three repeats. Hardware: Apple Silicon, CPU for classical/linear/correlation tiers, Apple MPS for deep and foundation tiers. External validation runs on AWS `r7i.8xlarge` through Open Problems' own Nextflow pipeline. Because wall-clock on a shared machine depends on co-scheduled load, external cost is reported as a ratio to a method present in every run. Environments are pinned and the Cell Ontology release is recorded.

Supplementary material (separate document) contains Figures S1–S4 — confusion matrices with ontology-equivalent errors outlined, and per-class recall for eleven methods on three splits — and Tables S1–S3.

Workflow components. A broad reference built from the CELLxGENE Census [CZI Census 2025]; a coarse→fine hierarchy obtained either from foundation-model embeddings or from Cell Ontology lineage; confidence-threshold abstention calibrated per reference by withholding 10% of cell types; masking-based refinement to a query's own supported classes or to a tissue; and a cluster-level novelty screen. Protocols for each are documented in the benchmark repository.

4 Results

4.1 Accuracy is clustered; cost is not

The top of the accuracy table is a four-way cluster spanning **0.008**, led by a tuned linear pipeline rather than by a deep model (Table 1). Those same four methods differ by **~260× in predict time** (0.34 s to 89 s) and **2.4× in peak memory**. actinn-jax holds the best ontology-aware concordance (0.811), a margin inside repeat noise and best read as a tie.

method	acc	macro-F1	ontology	fit (s)	predict (s)	peak mem (MB)
linear-anova-pca	**0.839**	0.699	0.808	**3.0**	**0.34**	4294
scANVI	0.833	0.697	0.809	0.0*	89.2	2099
scArches	0.832	**0.699**	0.805	54.7	21.4	1819
actinn-jax	0.831	0.683	**0.811**	21.9	**0.67**	2399
CellTypist	0.823	0.691	0.805	60.7	**0.66**	1569
SVM	0.810	0.680	0.797	66.5	**0.07**	1415
ProtoCloud	0.790	0.649	0.778	176.3	2.14	1893
SingleR	0.770	0.652	0.750	0.3	62.1	3541
kNN	0.770	0.622	0.769	0.4	**0.15**	1483
scTOP	0.739	0.619	0.703	1.1	1.07	1928
scmap-cluster	0.646	0.550	0.771	0.4	13.1	8073

Table 1. Accuracy and cost. Accuracy is the mean over the five shared-vocabulary datasets, macro-F1 over all six, and ontology concordance over the four that carry Cell Ontology ids; bold marks the best value in a column. *scANVI does most of its work in one train+predict pass, attributed to predict. Per-dataset scores: Supplementary Table S3.

No method leads everywhere. The linear pipeline and ProtoCloud each take two datasets, actinn-jax and scArches one apiece; actinn-jax leads the cross-study split, the regime closest to real reference mapping. The spread across leading methods on any one dataset is 1–4 points.

4.2 Rankings invert with reference size and with feature budget

Scaling the reference from 3k to 49k cells reverses the order: **ProtoCloud moves from worst (0.722) to best (0.976)**, clear of actinn-jax (0.936) and the linear pipeline (0.939), at 19× the CPU fit cost (Figure 1A). The HLiCA liver atlas reproduces the reversal independently (Figure 1B), and peak memory stays within a bounded band rather than widening (Figure 1C). Conclusions from subsampled references do not transfer to atlas scale in either direction.

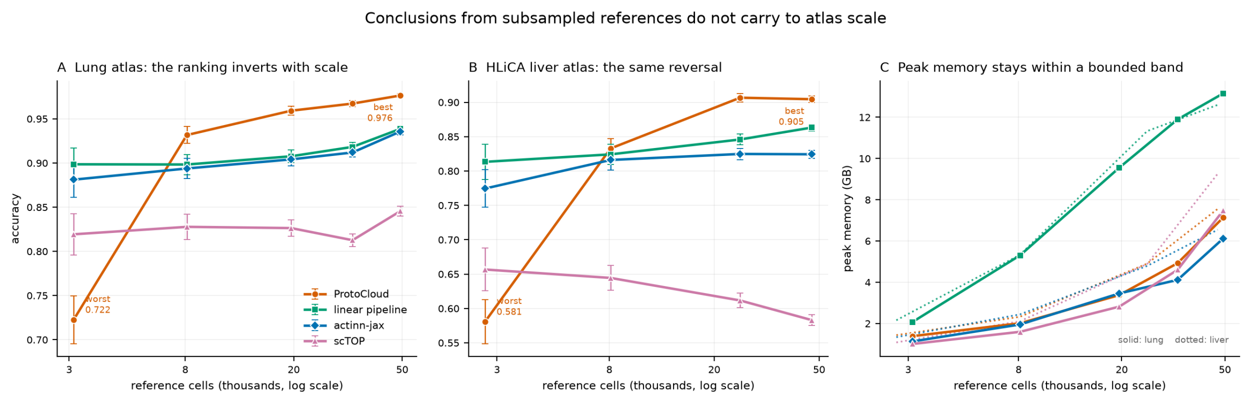

Figure 1. Accuracy and peak memory against reference size, four methods carried to atlas scale. *A:* lung, 3k → 49k reference cells. *B:* the HLiCA liver atlas, an independent replication. *C:* peak memory over both sweeps.

External validation inverts a different axis. On Open Problems — datasets, metrics and ranking not ours — all eleven methods were run in a single execution of its pipeline (Table 2). The tuned linear pipeline places third on accuracy and costs **2.7× more** than actinn-jax, having fit **7× faster** on our own panel. Open Problems supplies every method 1,000 highly variable genes, and an ANOVA→PCA→logistic pipeline pays for the decomposition on every query whereas a gene-space MLP amortizes it into a single fit. Neither a cost ranking nor an accuracy ranking survives a change of feature budget.

method	acc	macro-F1	cost	peak mem
mlp	0.843	0.662	1.98×	19.9 GB
actinn-jax	0.836	0.663	1.00×	21.0 GB
linear-anova-pca	0.828	0.647	2.67×	20.1 GB
SVM (SGD)	0.816	0.652	6.07×	20.1 GB
logistic_regression	0.813	**0.689**	**0.18**×	20.0 GB
CellTypist	0.810	0.643	7.62×	20.1 GB
knn	0.793	0.648	**0.07**×	19.5 GB

method	acc	macro-F1	cost	peak mem
xgboost	0.791	0.614	5.48×	80.7 GB
cellmapper_linear	0.776	0.553	0.35×	31.5 GB
naive_bayes	0.738	0.613	0.19×	19.5 GB
scTOP	0.581	0.462	**0.16**×	20.4 GB

Table 2. External validation on Open Problems `label_projection`, ordered by accuracy. Bold marks components we contributed to that benchmark. *cost* is per-dataset wall-clock relative to actinn-jax on the same instance. scTOP's mean reflects two collapses and one weak result inside three ordinary ones, traced to the fixed feature budget.

4.3 Inference cost is flat in reference size

Fit time grows with reference size and cardinality for every trained method. Prediction does not: for a cached reference model it stays **0.08–0.33 s whether the reference holds 1k or 24k cells, and 5 or 86 types** (Figure 2). With train-once/map-many caching the fit is paid once and only the flat prediction cost recurs — the regime that matters when one reference serves many queries. Peak memory is likewise bounded rather than divergent: the linear/actinn-jax ratio holds at ~2–3× (6.1 vs 13.2 GB at 49k cells) rather than widening (Figure 1C). The same flatness holds on the query axis: with the reference fixed, actinn-jax annotates the entire 524,699-cell liver atlas in **57.9 s** at 15.0 GB and 6,100–9,100 cells/s, while the tuned linear pipeline's throughput falls with query size and its full-atlas run did not finish.

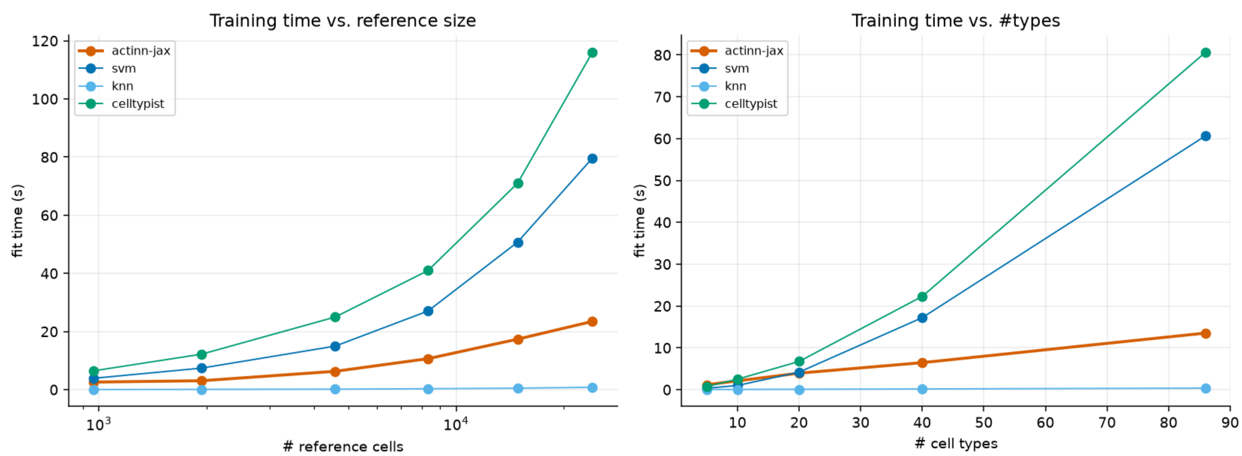

Figure 2. Fit and prediction time against reference size and label cardinality. Fit time grows for every trained method; prediction stays flat and sub-second across the whole range.

4.4 A flat cost profile makes multi-stage annotation practical

Because each stage is a cached model with sub-second, memory-bounded inference, several can be chained on a laptop (Figure 3). A shipped ~800-type census reference gives any query a first-pass annotation with calibrated abstention; a small focused reference then re-annotates at full resolution. On withheld cross-study liver cells the broad pass scores **0.23 exact / 0.58 ontology** and the focused reference reaches **0.72 / 0.86** on the same cells.

The two passes hand off; they do not combine. Substituting a stronger broad model lifts the broad call but changes nothing downstream, and using it to narrow the focused pass's classes makes the result **worse** (0.731 → 0.708), because a wrong mask discards the correct class outright. Once the focused reference covers the tissue, the broad model's value is routing, not resolution.

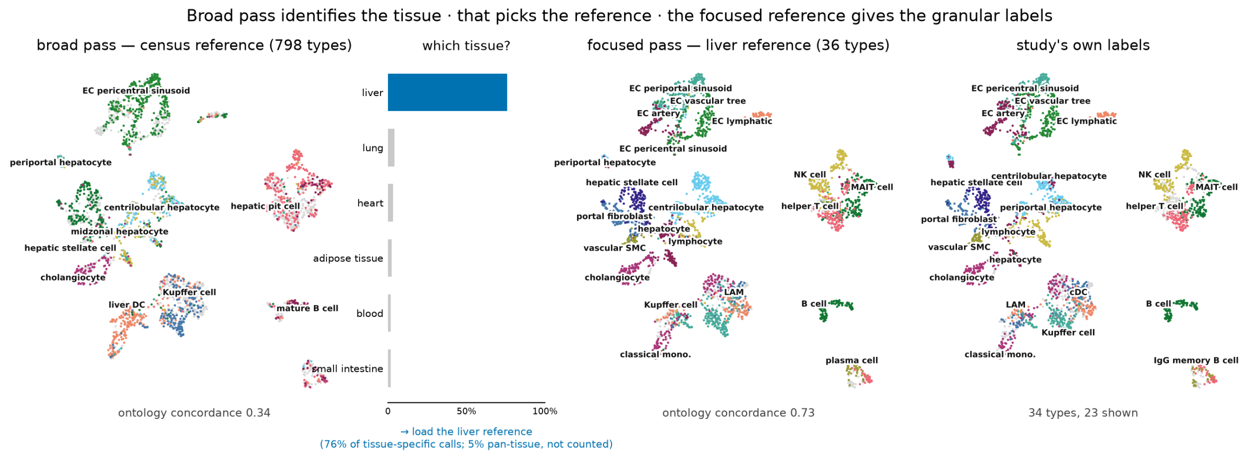

Figure 3. The workflow on a withheld liver study (3,396 cells), same embedding throughout. The census reference spreads 137 of its 798 labels over the query (concordance 0.34); resolving those calls to tissue gives **76% liver** against 4% for the next candidate, which selects the reference to load; the 36-type liver reference then re-annotates the same cells at **0.73**, tracking the clusters. Rightmost panel is the study's own labels.

4.5 A pretrained annotator can be distilled without a GPU or labels

Building the broad pass from the census requires a foundation model on a GPU to discover its hierarchy. A pretrained pan-human annotator [Sarkar et al. 2026] already publishes one, so labeling a corpus with it and training on those labels transfers both vocabulary and structure — using **only raw counts**. On withheld cross-study liver cells the distilled model **matches the teacher and beats the census-built reference**, at several times the teacher's throughput, in under ten minutes of CPU (Table 3).

broad-pass model	classes	ontology	cells/s
census-built, ours	798	0.338	2,962
Pan-human Azimuth (teacher)	382	0.380	1,076–1,563
distilled from Azimuth, ours	324	**0.406**	**8,937–10,021**

Table 3. Broad-pass entry points on 3,396 withheld cross-study liver cells. The distilled student needs only raw human counts to build — no GPU, no labeled input — and inherits the teacher's vocabulary and hierarchy. It does not inherit the teacher's calibrated abstention, which remains a limitation. The 0.406/0.380 ordering is one query, and both actinn-jax models draw on a census sample that may include these studies while Azimuth does not, so read the student as level with its teacher rather than ahead of it.

The GPU can be removed from the other route as well. Deriving the coarse hierarchy from Cell Ontology lineage rather than from a foundation-model embedding beats both no hierarchy at all

(0.616 vs 0.547) and a same-sized random grouping (0.539), so it is the lineage structure and not the mere act of splitting that helps. Being species-independent, that route also reaches a second organism: a pan-mouse reference of 453 types across 85 tissues reaches 0.638 ontology concordance on two withheld datasets after 17 s of CPU.

4.6 Abstention is tunable; zero-shot labels are not competitive

Withholding 9 of 36 cell types entirely and sweeping a confidence threshold over the eight methods that return a per-cell confidence, five trade coverage for accuracy across the range and three do not: CellTypist's probabilities are saturated and give one operating point instead of a curve, scTOP's projection score keeps only 6% of the query at $p \geq 0.5$, and ProtoCloud's ambiguity flag is flat until 0.9. Among the five that work, abstain quality does not separate them — actinn-jax and scArches are effectively tied (0.969 accuracy on 66% of cells with 73% of novel cells flagged, against 0.983 on 61% with 71%) — but cost does, at 0.67 s of predict time against 21.4 s. Separately, a foundation model run zero-shot scored **0.201** ontology concordance against 0.917 for a reference-trained model on the same cells, taking ~280× longer — reproduced independently on Open Problems, where zero-shot entries sit at the bottom of the leaderboard.

5 Discussion

Across two independent benchmarks the same pattern holds: methods separated by less than a percentage point of accuracy are separated by two orders of magnitude in cost, and neither ordering is stable across reference size or feature budget. Reporting accuracy alone therefore under-determines the choice of method, and reporting it from a single reference size can invert the recommendation.

Two consequences follow. First, benchmark design should treat reference size and input budget as axes rather than as fixed settings; our own results reverse along both. Second, once several methods are accurate and cheap, the productive contribution is not a further ranking but a characterization of fit — which is why we report accuracy-per-second and accuracy-per-byte, and demonstrate what a low flat cost profile enables rather than arguing from the accuracy column.

We are explicit about what is not ours. ProtoCloud provides uncertainty, attribution and a retraining-based refinement path, and becomes the most accurate method at atlas scale. A purpose-built pan-human annotator is better resourced than our census-built reference and we make no claim to improve on its annotations; we distill it. What remains distinct is the hand-off no fixed typology performs — re-annotation into a user's own label set, resolution below a typology's leaves, and screening for what none of them claims — at a cost that keeps every stage on the machine already on the desk.

The main limitations are that the cross-method comparison is human-only and single hardware-family; that our classical tier is untuned while the linear baseline is tuned, which biases against our own method; and that the distilled reference inherits a vocabulary but not its teacher's calibrated abstention. Full limitations are in the extended report.

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